

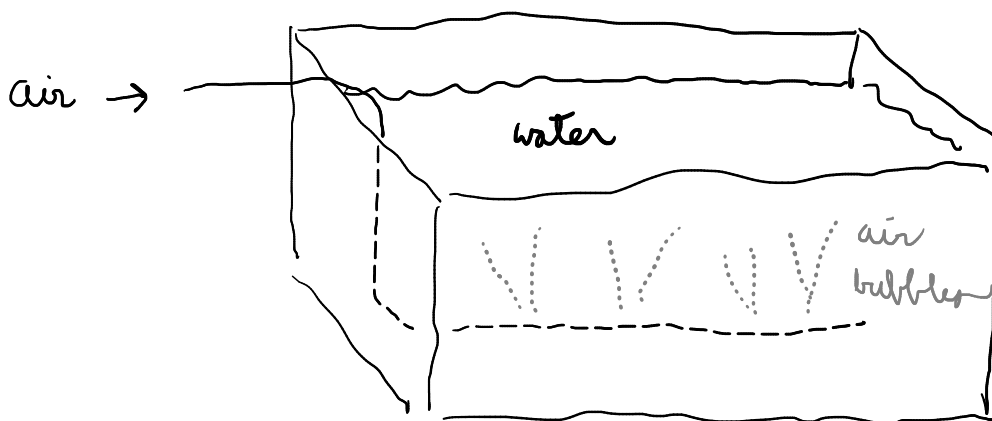
Aeration solution Modelling 2025

Model formulation steps

Formulating a model includes several steps that can be organized like this:

1. Initial concepts and assumptions (boundary, state variables, sketches)
2. Balance/conservation equations
3. Constitutive equations (transfer or rate equations)
4. Linking equations
5. Spatial or other simplifications
6. Initial or boundary conditions
7. Governing equation
8. Analytical or numerical solution

1.



Our state variable is the dissolved oxygen mass or concentration in the tank water.

Let's assume the tank is well-mixed, so dissolved oxygen concentration is uniform. So this will be a lumped-parameter model. We do not have to try to simulate how oxygen concentration varies over space or how the O₂ concentration changes in bubbles as they rise. Will our solution be accurate? Well, the tank should in fact be very well mixed.

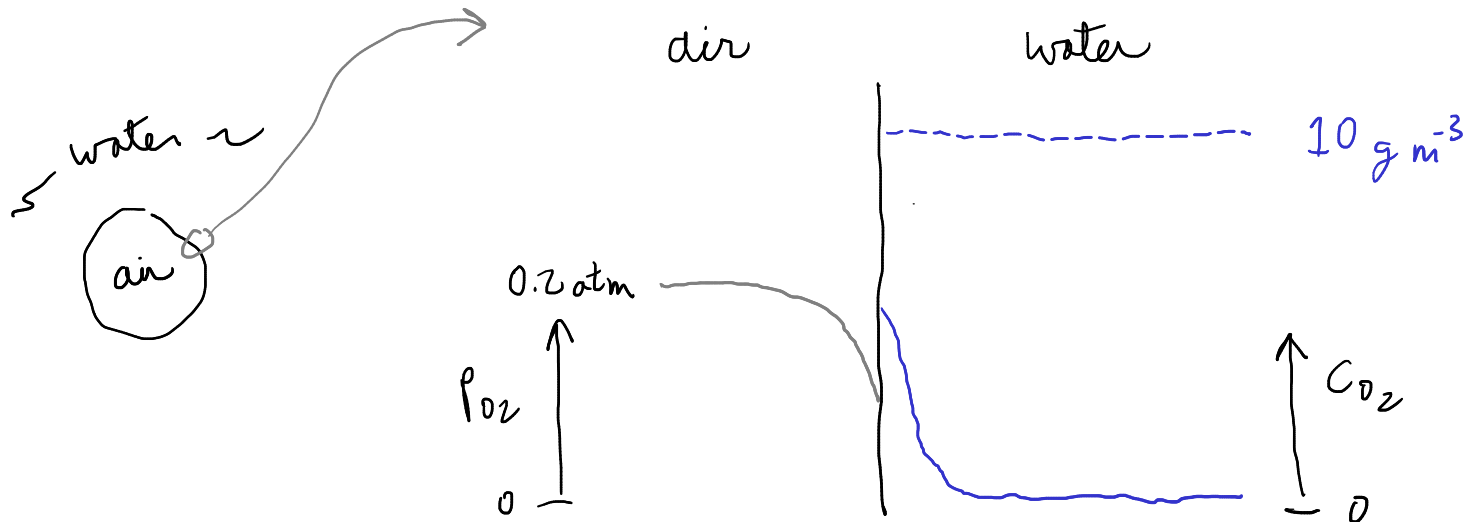
And the application of a $k_L a$ value implies that oxygen delivery is never limiting (mass transfer is). So we don't have to explicitly consider how much oxygen we deliver. Of course it would have to be sufficient! To attain the specified $k_L a$ we need to provide a sufficient flow of air with sufficiently small bubbles, and application of the $k_L a$ implies that this is the case.

How about our model boundary? We would typically use the tank or water boundary. That is fine. But if so we need to remember it is the *dissolved* oxygen content that is the state variable. Oxygen in the bubbles is not part of this. Alternatively, we might say that the boundary is around just the water, and it excludes the bubbles. Good luck drawing that!

Do we need to consider oxygen loss from the water? A few of you asked about this when you were working on the exercise. No, we do not. Mass transfer constitutive equations are typically for *net* mass transfer. Sure, there is some flow of oxygen in both directions but we are interested in the net, measurable flow.

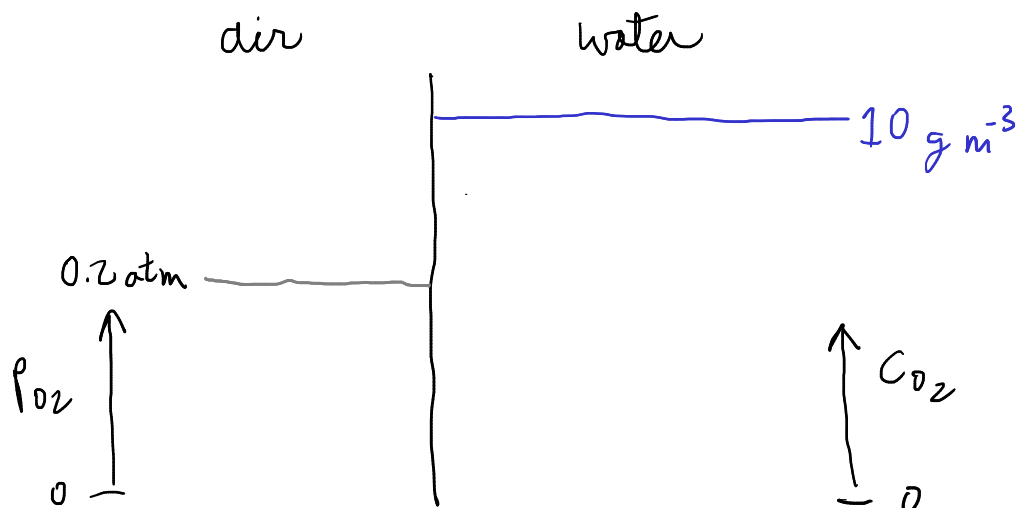
Because interphase mass transfer is complicated, I will explain a bit more about our conceptual model than usual here. But I do not expect you to provide this level of detail in your solutions!

At the start of aeration, the volume around the typical or average air bubble/water interface would look like this:

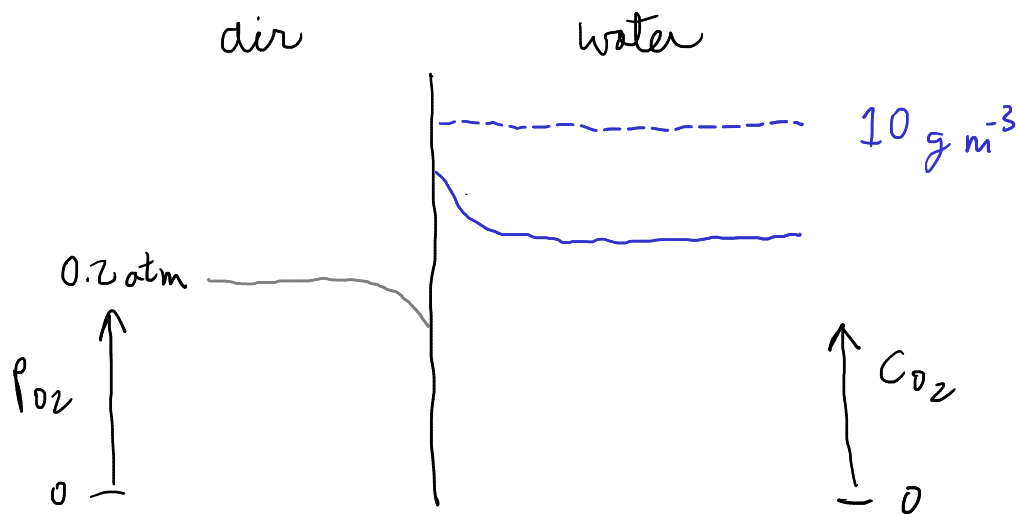


So we have (net) oxygen transfer from the bubble to the water. Remember that our concentration (or partial pressure) scales on either side of the interface cannot be directly compared! Instead, we have to convert the air phase concentration or partial pressure to an equivalent water phase value using Henry's law constant. That was already done, and the exercise description gives you a saturation concentration (called c_{sat} or c^*) $10 \text{ mg/L} = 10 \text{ g/m}^3$, and that is shown as the dashed line above.

Because we have no consumption (at least in the first part of the exercise) we expect the system to eventually come to equilibrium, instead of steady state, although the distinction can be a bit arbitrary in my opinion. At that point our profiles would look like this:



How about with oxygen consumption in the water? That would maintain an oxygen concentration in the water below the saturation value.



Exactly how far below depends on the consumption rate and $k_L a$.

You might ask if it is reasonable to assume a uniform concentration of oxygen within the water (shouldn't it be higher near the surface because bubbles rising have given up more oxygen?) and a uniform and constant oxygen concentration (or partial pressure) within the bubbles (based on the same reasoning, shouldn't bubbles by the surface have less oxygen, and also bubbles at the start have less oxygen, because they give up more?). The answers to the questions in parentheses are yes in all cases. But we are ignoring these effects in our simple model, arguably because we suspect that the impact of these effects on predictions are small, especially compared to uncertainty in our mass transfer coefficient, but mainly because the lumped parameter simplification makes it much easier to solve the model! Ultimately we should evaluate the simple model before we trust it, and in this case that could be done by looking at all the studies on water aeration that show this simple approach is reasonably accurate and very useful.

One last concept to think about here in the first step. . . although it was explained a couple times already. But it is potentially confusing. A mass transfer coefficient times a concentration difference gives us a flux, like this

$$J = k_L (C_{sat} - C_w) \quad (g \cdot m^{-2} \cdot h^{-1})$$

To get to flow rate (g oxygen per hour) we should multiply by the bubble area, because that is the area of the interface over which mass transfer occurs. In tank aeration, this area is typically expressed as a volumetric value (m^2 area per m^3 tank or water volume) and lumped in with k_L and written as $k_L a$, i.e., the parameter that has been provided.

Now, do we really know the value of k_L and the separate value of specific area a ? Nope! We just have the lumped value, but do not need to think about the two separately so it does not matter.

2.

Accumulation rate = oxygen rate in - oxygen loss + oxygen generation - oxygen consumption

Here we'll assume that there is no loss of oxygen, no generation, and no consumption.
So we get:

Accumulation rate = oxygen transfer in, from bubbles to water

$$g \text{ s}^{-1} \quad g \text{ s}^{-1}$$

Now, our kLa is already per hour so let's use $g \text{ h}^{-1}$

And it is already a volumetric term, so let's normalize our mass balance by tank volume and actually use

$$g \text{ m}^{-3} \text{ h}^{-1}$$

This type of normalization is an important form of model simplification that can make model development and application to other systems easier.

And, to make a later step easier, instead of "accumulation" we will write

$$\frac{dC_{O_2}}{dt} = \text{volumetric } O_2 \text{ mass transfer rate} \quad (g \text{ m}^{-3} \text{ h}^{-1})$$

I have used a few different terms for the rate of oxygen transfer into the water. Sorry about that. But maybe it is useful to see the equivalence.

3.

$$\frac{\dot{m}}{V} = k_L a (c_{\text{sat}} - c_t)$$

saturation O_2 conc.

volumetric O_2

mass transfer
rate

$$(g_{O_2} m_{H_2O}^{-3} h^{-1})$$

RHS of our mass balance
equation from 2

4. None needed.

5. Nothing spatial here. We are assuming uniform concentrations. That is not a bad assumption for an agitated tank.

6. Initial dissolved oxygen concentration is zero.

$$c_t(0) = 0$$

7. Put 2 and 3 together.

c_t = dissolved O_2 conc.
in tank ("t")

$$\frac{dc_t}{dt} = k_L a (c_{\text{sat}} - c_t)$$

8. We will solve this numerically using Python. See `aeration_num.py` and `aeration_mods.py` etc for details on exactly how it is done. Here I will present a description. And then below I will solve it analytically.

For a numerical solution, we will use the finite difference method. With that we do not need to explicitly solve that governing equation above symbolically. We just have Python evaluate it repeatedly, updating the state variable value (c_t) each time.

The simplest way to do this is with Euler's method, using an explicit forward approach. We can do that in essentially two steps in Python: 1) create an array to hold our results, with one element for each time we want in the output, setting the first value to the initial dissolved oxygen concentration, and 2) loop through the elements, changing the value of the state variable by $+ dc/dt * dt$ each time. In Python this looks like:

```
for i in range(1, len(conc_o2)):
    conc_o2[i] = conc_o2[i - 1] + kla * (csat - conc_o2[i - 1]) * dt
```

This simple approach can be surprisingly effective, but it can fail spectacularly if the time step dt is too large. While you can go far with it by simply selecting a good time step, there are more accurate and definitely much faster approaches. A better option is to use the `solve_ivp()` function, which uses a 4th or 5th order finite difference approach, considering the derivative not just at fixed time steps, but before and after as well. Ask Frederik for details :). For better or worse, we can apply it without really understanding the algorithm details, other than knowing it calls up some function for calculating derivatives repeatedly, and updates the state variable value as it goes. That looks like this in Python:

```
# Try a more sophisticated approach--solve_ivp()
def rates(t, c):
    return kla * (csat - c)

res = solve_ivp(rates, t_span = (0, 1), y0 = [0], t_eval = times)
```

I named the function that calculates the value of the derivative `rates()`

See the posted Python scripts for more details.

An analytical solution is a symbolic solution. For that we need to know something about solving ODEs symbolically. Fortunately, this ODE has the form (or can be simplified to have the form) of the one we know how to solve.

← our G E

$$\frac{dc_t}{dt} = k_L a \cdot (c_{sat} - c_t)$$

If we can get it into the form of the ODE in the exp_decay_growth... document, like this

$$\frac{dx}{dt} = k \cdot x$$

Then we could easily solve it, with a solution that has this form

$$x = x_0 \cdot e^{kt}$$

So how do we do that? We can replace the concentration difference with a symbol for a new variable delta c, which is often called the "oxygen deficit". This is good practice anyway, because it gives us a more general model that could be applied to any starting concentration or saturation concentration without making changes.

$$\Delta c = c_{sat} - c_t \rightarrow c_t = c_{sat} - \Delta c$$

$$\frac{d \Delta c}{dt} = k_L a \Delta c$$

A

But then we need an additional step, because for this to be an (easily interpreted) ODE and certainly for it to have the form we want, the two parts circled above must be the same variable.

So from this and this above, we can substitute A into B and write

$$\frac{d(c_{sat} - \Delta c)}{dt} = k_L a \Delta c$$

And the LHS simplifies to (because addition of a constant term like c_{sat} does not affect a derivative, and the negative sign - is a constant -1 multiplied, which can be brought to the outside and then the RHS for

$$\frac{d\Delta c}{dt} = -k_L a \Delta c$$

This has the form we want! So a solution is:

$$\Delta c = \Delta c_0 e^{-k_L a \cdot t}$$

Or, if we prefer an expression for c_t

$$c_{sat} - c_t = (c_{sat} - c_{t0}) e^{-k_L a \cdot t}$$

And because we know that $c_{t0} = 0$ (see initial value step above), this simplifies to

$$-c_t = c_{sat} e^{-k_L a \cdot t} - c_{sat}$$

Which further simplifies to

$$c_t = (1 - e^{-k_L a \cdot t}) c_{sat}$$

Implemented in Python, this solution might look like this:

```
# Results array
conc_o2 = csat * (1 - np.exp(-kla * times))
```

where `times` is a NumPy array of times for evaluation. Analytical solutions are very easy to implement (program) once you have them!

And, importantly, you can infer things about a system's behavior by simply looking at an analytical solution. In this case, if you know what the mathematical function $\exp(x)$ does as x goes from 0 to infinity, you can see just from looking at this solution

$$c_t = (1 - e^{-k_L a \cdot t}) c_{sat}$$

that c_t will start at 0, increase rapidly, and then more and more slowly over time as it approaches (but never reaches) c_{sat} . That is the kind of useful inference you can make from analytical solutions. Typically you cannot do this with a numerical solution without actually solving it, and for complex models with a lot of parameters, solving it many times with different input values.

If our system had a fixed rate of oxygen consumption in the tank, we would need to go back to the mass balance step (2), and get:

Accumulation rate = oxygen transfer in - oxygen consumption rate

$$\text{all } g_m^{-3} h^{-1}$$

$$\frac{d c_{O_2}}{d t} = Q_{O_2 \text{ transfer in}} - Q_{O_2 \text{ consumption}}$$

Then for our constitutive equations (3), we still have mass transfer

$$\frac{\dot{m}_{aeration}}{V} = k_L a (c_{sat} - c_t)$$

$\swarrow$
 volumetric O_2
 mass transfer
 rate

But we also have a some fixed rate of consumption

$$\frac{\dot{m}_{\text{cons}}}{V} = r_{m, \text{cons}}$$

So for step 7 we have

$$\frac{dc_t}{dt} = k_L a (c_{\text{sat}} - c_t) - r_{m, \text{cons}}$$

our new G E

Evaluate at steady state :

$$\frac{dc_t}{dt} = 0$$

$$k_L a (c_{\text{sat}} - c) = r_{m, \text{cons}}$$

$$c_t = c_{\text{sat}} - \frac{r_{m, \text{cons}}}{k_L a} \text{ at steady state only!}$$

If you are good at symbolic math and solving ODEs, you could also come up with an analytical solution for c (c_t) over time. In contrast, you do not need to be good at math at all to implement consumption in a numeric model! Just add it in to the derivative function, following the new governing equation above. In Python this could be done like this:

```
# Define derivatives function
def rates(t, cc):
    return kla * (csat - cc) - cons
```